

Assignment - 2

classmate

Date _____

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Web Services.

Q1] Attempt all questions.

1] What is soap?

⇒ SOAP (Simple object Access protocol) is a messaging protocol specification for exchanging.

2] Write a note on SOAP communication model.

⇒ The SOAP communication model typically follows a request - Response pattern. It involves three major roles.

- Service Provider - The entity that ~~access~~ creates the web service & make it available.
- Service Requestor - The client applⁿ that needs to use the service.
- Service Registry - A central directory where providers publish descriptions so requestors can find them.

A SOAP message is bundled in an Envelope which contains a Header & a body.

3] Write a note on WSDL

⇒ WSDL (web services Description Language) is an XML-based file that tells the client application what the web services does & how to connect to it.

- The "Contract": It acts as a contract both the provider & the consumer.
- Content: It defines the services location (URI), the available methods. The data types

used, & the specific message formats required for communication.

Q Write a note on UDDI

→ UDDI (Universal Description, Discovery & Integration) is an XML-based registry for businesses worldwide to list themselves on the internet.

- The "Yellow Pages": Think of it as a phone book for web services.
- Functions: It allows businesses to discover each other and define how they interact over the internet. It works in conjunction with SOAP & WSDL.

Q Explain REST Architecture

→ REST (Representational State Transfer) is an architectural style for designing networked applications. It is often preferred over SOAP for web-based services because it is lighter & faster.

- Stateless: Each request from a client to a server must contain all the information needed to understand & complete the request.
- Resources - Everything is considered a "resource" identified by a URL (Uniform Resource Identifier)
- HTTP methods - It uses standard HTTP verbs like GET (read), POST (create), PUT (update), & DELETE (remove)

Q Explain JSON

⇒ JSON (Javascript Object Notation) is a lightweight data-interchange format that is easy for humans to read/write & easy for machines to parse/generate.

- Language Independent -

Although derived from JS it is supported by almost all modern programming languages.

- Format - Data is represented as key-value pairs (eg. "name": "John") & Arrays. It is the primary data format used in RESTful APIs because it is less verbose than XML.

Q Write SOAP Example (Program)

⇒ A SOAP request is a specific XML structure

XML

```
<?xml version="1.0"?>
<soap:Envelope xmlns:soap="http://www.w3.org/2003/11/soap" >
  <soap:Header>
  </soap:Header>
  <soap:Body>
    <m:GetStockPrice xmlns:m="http://www.example.org">
      <m:stockName>GOOGLE</m:stockName>
    </m:GetStockPrice>
  </soap:Body>
</soap:Envelope>
```